Arroyo et al's model with a high-temperature cutoff

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Let's start from Arroyo et al (2022) original model, expressed using (non dimensional) inverse temperature $\beta = T_0/T$ (recall that $T_0 = 273K$). The model reads

$$k(T) = k_0 T^{-a} \exp\left(-\frac{\Delta H}{k_B T}\right) \propto \beta^a e^{-\beta/\epsilon};\ \beta = \frac{T_0}{T} = \frac{1}{1+\tau};\ \epsilon = \frac{k_B T_K}{\Delta H}$$

In this model, optimal inverse temperature is simply

$$\beta_* = a\epsilon$$

So me might as well reparametrize the model as (replacing ϵ by β_*/a)

$$RP_f(\beta) = \frac{k(\beta)}{k(\beta_*)} = \left(\frac{\beta}{\beta_*} e^{1-\frac{\beta}{\beta_*}}\right)^a$$

To see why we get a gaussian limit for large large a, we can first note that if a is large only terms close to 1 in the parenthesis will contribute. This suggests to center around β_*, defining $z = 1 - \frac{\beta}{\beta_*}$, (which in temperature variables is $z = \frac{\tau - \tau_*}{1+\tau} \approx \tau - \tau_*$). This leads to

$$(((1-z)e^z))^a = \exp\left(a\log(1-z) + az\right) = \exp\left(-a\frac{z^2}{2} - a\frac{z^3}{3} - ...\right)$$

given that the relevant scale of observation of z is bounded, we get that the function converges to a gaussian with variance $1/a$. To be clear the model does not always produce a gaussian TPC, it can indeed be skewed. **However, it will always be skewed in the opposite direction as what is usually found in data** (see Fig 1 left panel). To solve this issue, we can introduce a smooth cutoff function χ that vanishes at β_c, whose mechanistic origins may be unknown. Here, just to fix ideas we suppose that it comes from the fact that performance is really a difference of two rates (as in the trade-off model that we proposed in the SI, but we could redo the same analysis with a general function $\chi(\beta - \beta_c)$). So let us set:

$$\chi(\beta - \beta_c) = (1 - e^{-\frac{\beta - \beta_c}{\delta}});\ \beta > \beta_c$$

But its precise form may not matter that much. Performance is now:

$$k(\beta) \propto \chi(\beta - \beta_c) \times \beta^a e^{-\beta/\epsilon}$$

optimal inverse temperature satisfies

$$\frac{\epsilon a}{\beta_*} + \epsilon \frac{\chi'(\beta_* - \beta_c)}{\chi(\beta_* - \beta_c)} = 1$$

which for the simple cutoff function we chose can be rearranged to read:

$$e^{\frac{\beta_* - \beta_c}{\delta}} = 1 + \frac{\epsilon}{\delta}\left(1 - \frac{\epsilon a}{\beta_*}\right)^{-1}$$

We now set

$$\hat{x} = \frac{\beta_* - \beta}{\epsilon} = \beta_* x \beta;\ x = \frac{\tau - \tau_*}{\epsilon}$$

and note that $\hat{x} \approx x$ (the rescaled variable from the main text) as long as β stays close to 1, which is the case when considering normal temperatures, suitable for life. Rescaled and recentered performance y now reads

$$y(\hat{x}) = e^{\hat{x}}\left(1 - \frac{\epsilon a}{\beta_*}\frac{\hat{x}}{a}\right)^a \left(1 - \left(1 - \frac{\epsilon a}{\beta_*}\right)\frac{e^{\frac{\epsilon}{\delta}\hat{x}} - 1}{\epsilon/\delta}\right)$$

if ϵ/δ is small enough, the details of the cutoff will not matter as $\frac{\delta}{\epsilon}(e^{\frac{\epsilon}{\delta}\hat{x}} - 1) \approx \hat{x}$ and **only the effect of the power law structure remain**s:

$$y(\hat{x}) \approx e^{\hat{x}}\left(1 - \frac{\epsilon a}{\beta_*}\frac{\hat{x}}{a}\right)^a \left(1 - \left(1 - \frac{\epsilon a}{\beta_*}\right)\hat{x}\right)$$

This model –**which now indeed converges to UTPC** if $\frac{\epsilon a}{\beta_*}$ small– can be fitted to data to see how deviations from UTPC could inform on the value of the exponent a. As we see in the figure below, this effect can be detectable if the cutoff value β_c is large. **Providing a clear rigorous road map to systematically explore deviations from UTPC constitutes an exciting line of future enquiry** (We thank Samraat Pawar for this suggestion).

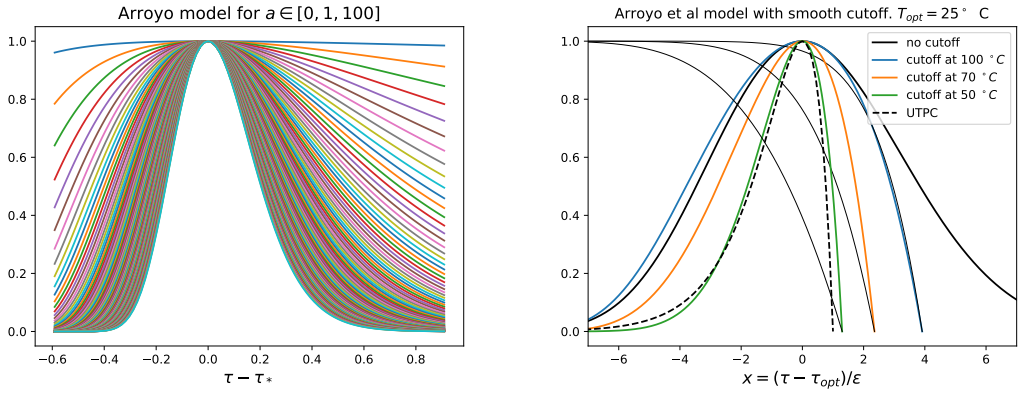

Figure 1: **Left:** TPC based on Arroyo et al's model for values of the parameter a, for large $a$we see the gaussian limit. For small a we get right-skewed TPCs. **Right:** Rescaled TPC based on Arroyo et al's model with a smooth cutoff (grey lines). We modify the value of the cutoff $\beta_c = T_0/T_c$ and adjust a so that the optimal temperature is arbitrarily fixed at 25 degree Celsius. Note that in the absence of a cutoff the TPC is almost gaussian (thus symmetric). This a major deviation from UTPC, but this deviation vanishes as the cutoff value is reduced. Thus, the effect of the power law structure predicted by Arroyo et al's theory requires this cutoff value to occur at relatively large temperatures.